

Technology Stack

Digitize Metro Ticket Booking Using ServiceNow

Date	2 August 2026	Team ID	Not Specified
Project Name	Digitize Metro Ticket Booking Using ServiceNow	Maximum Marks	4 Marks

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Purpose: Define the technologies and ServiceNow components required to deliver a digital metro ticket booking workflow.

Layer / Component	Technology / Service	Purpose
Platform	ServiceNow	Central platform for ticket booking requests, workflow, records and administration.
User Interface	Service Catalog / Service Portal	Passengers select source station, destination, travel date and ticket quantity.
Data Layer	Custom ServiceNow tables	Stores booking details, fare information, ticket status and QR token.
Automation	Flow Designer	Validates booking data, calculates fare, creates ticket record and triggers notifications.
Business Logic	Business Rules / Script Includes	Supports validation, fare rules and reusable server-side logic where required.
Notifications	ServiceNow Notifications	Sends booking confirmation, ticket details and status updates.
Digital Ticket	QR Code generation service / integration	Creates a QR representation of the booking reference for validation.
Security	Roles, ACLs and authentication	Protects passenger and booking information and restricts administrative access.
Reporting	ServiceNow Reports / Dashboards	Tracks booking volume, routes, failures, peak demand and operational trends.
Integration	REST APIs / Integration Hub (optional)	Enables future integration with payment gateways, metro systems or external validation services.

Proposed Architecture

Passenger → Service Catalog → Booking Record → Flow Designer → Fare Calculation → QR Ticket Generation → Notification → Reporting Dashboard

Technology Selection Rationale

ServiceNow is selected because it combines a user-facing catalog, structured data, workflow automation, notifications, security and reporting in one platform. This reduces manual handoffs and provides a foundation for future integrations and operational monitoring.

Client-Side Technologies

Catalog Client Scripts

Used for

- Validating station selection
- Checking passenger count
- Calculating fare
- Preventing invalid submissions

UI Policies

Used for

- Making fields mandatory
- Hiding unnecessary fields
- Showing Return Date only when Return Journey is selected
- Making fields read-only

UI Actions

Provide custom buttons such as

- Generate QR Ticket
- Cancel Booking
- View Booking

Server-Side Technologies

Business Rules

Business Rules automatically execute whenever booking records are inserted or updated.

Functions include:

- Booking validation
- Fare calculation
- Booking status update
- QR code generation trigger
- Database updates

Script Includes

Script Includes store reusable business logic.

Examples

- Fare Calculation Function
- QR Code Generator Function
- Ticket Validation Function
- Passenger Verification Function